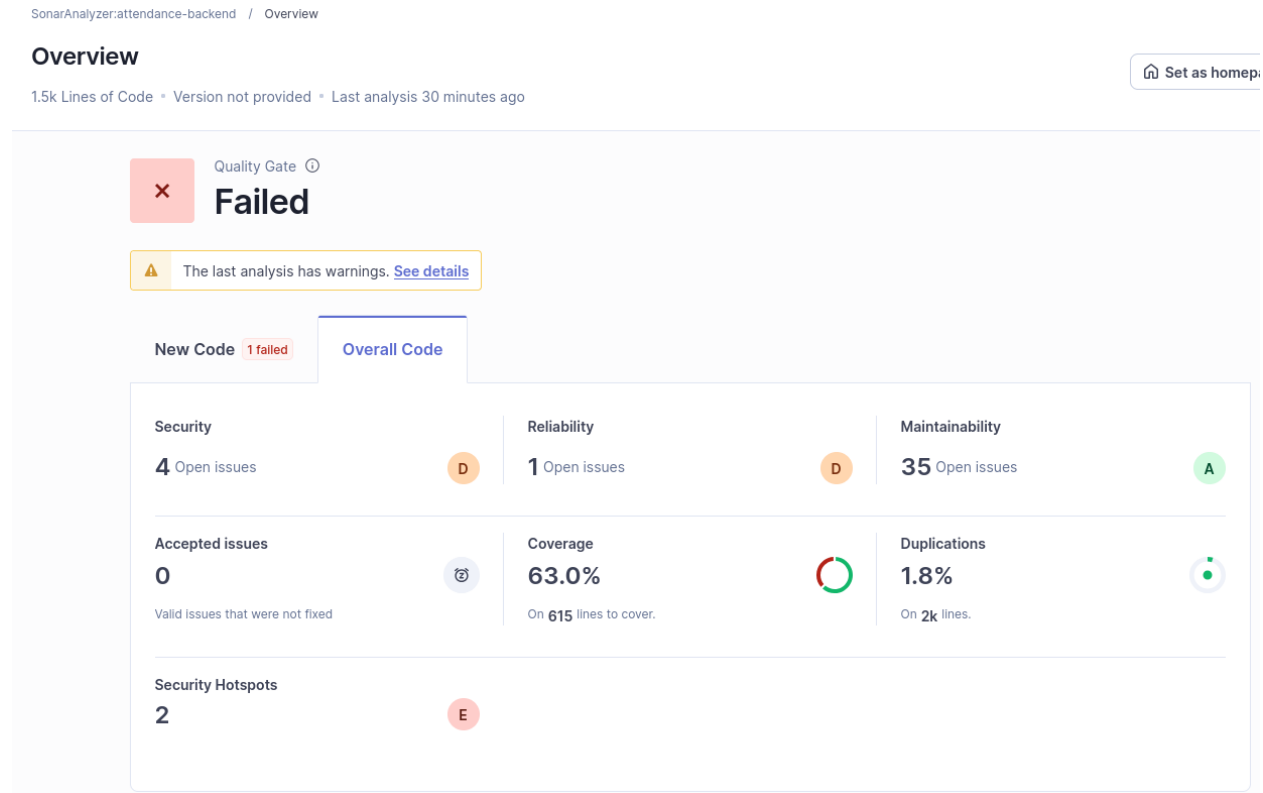


For static code analysis, we used SonarQube. The first overview:



In Measures Project Overview we can see that crucial issues in the code are present in LectureController.java and CorseController.java classes. We added tasks for our project to fix them.



The issue of Reliability was very simple and fast fixed.

Issues

My Issues

All

Filters

Clear All Filters

Looking for Bugs, Vulnerabilities, or Code Smells? If your team prefers working with these types, change it in the [settings](#)

Issues in new code

Software Quality ⓘ 1 X

Security 4

Reliability 1

Maintainability 35

Add to selection Ctrl + click

Severity ⓘ

Blocker 0

High 1

Medium 0

Low 0

Info 0



Bulk Change

Select issues ▲ ▼

Navigate to issue ▾

src/.../attendancebackend/service/LectureService.java



Save and re-use this "Random".

Reliability High ▼

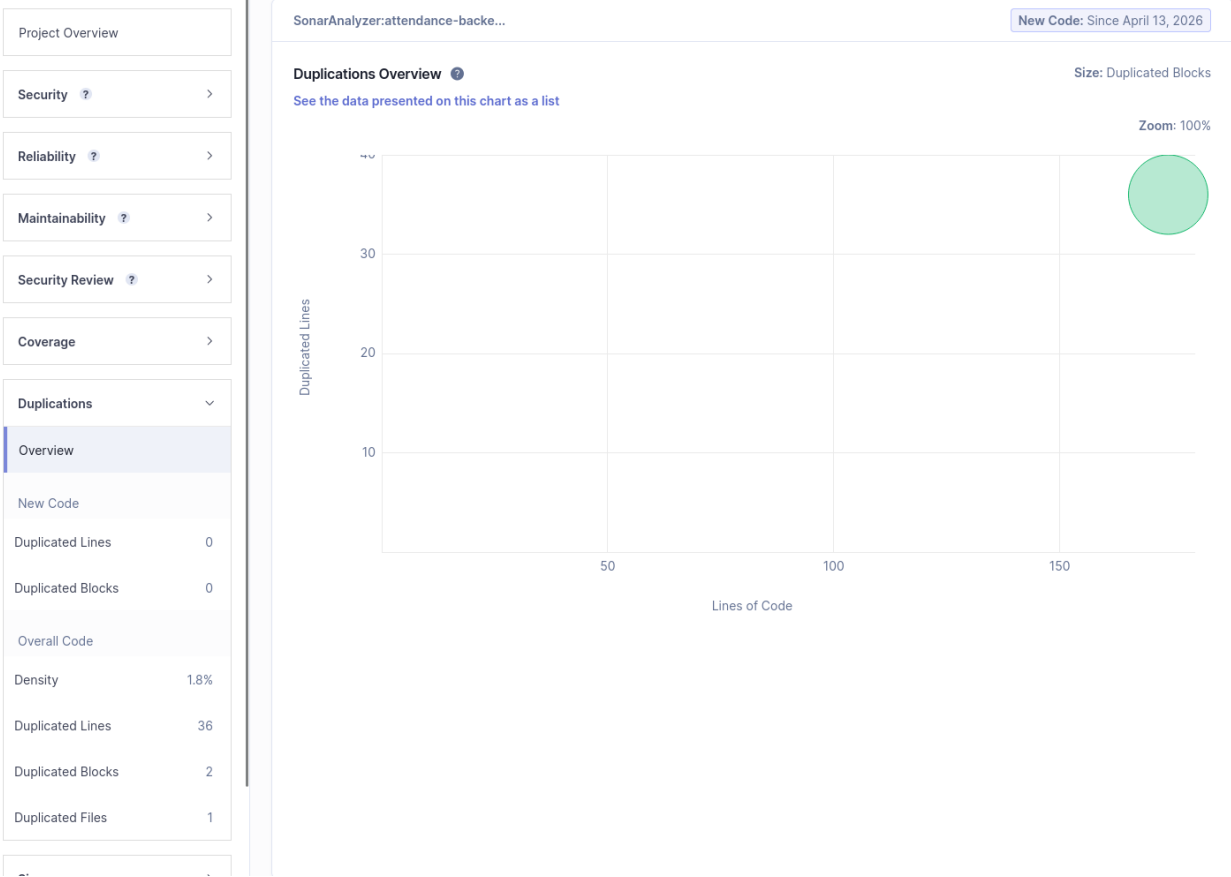
Open ▼

Not assigned ▼

1 of 1 shown

Some Code Metrics:

Duplicated lines are present only in StudentDashboardService.java.



Cyclomatic complexity of the whole project is 225 across 107 methods.

Project Overview

Security ? >

Reliability ? >

Maintainability ? >

Security Review ? >

Coverage >

Duplications >

Size >

Complexity ? >

Cyclomatic Complexity 225

Cognitive Complexity 125

Issues >

src/main > java/org/example/attendancebackend View as Tree 8 items

Cyclomatic Complexity 225

config	15
controller	54
dto	0
entity	10
repository	0
service	134
util	11
AttendanceBackendApplication.java	1

8 of 8 shown

The biggest part of cyclomatic complexity is concentrated in the service layer. This means that most of the branching logic in our application, such as if-else conditions, validation checks, and error handling, is concentrated in the service classes.

```

109 public void applyTranslationsBatch(List<Course> courses, String requestedLocale) {
110     if (courses == null || courses.isEmpty()) {
111         return;
112     }
113     String wanted = LocaleUtil.normalize(requestedLocale);
114     List<Long> ids = courses.stream().map(Course::getId).filter(Objects::nonNull).toList();
115     List<CourseTranslation> all = translationRepository.findByCourseIdIn(ids);
116     Map<Long, List<CourseTranslation>> byCourse = new HashMap<>();
117     for (CourseTranslation t : all) {
118         byCourse.computeIfAbsent(t.getCourseId(), k -> new ArrayList<>()).add(t);
119     }
120     for (Course c : courses) {
121         pickTranslation(c, byCourse.getOrDefault(c.getId(), List.of()), wanted);
122     }
123 }
124
125 private static void pickTranslation(Course course, List<CourseTranslation> rows, String requestedLocale) {
126     if (rows.isEmpty()) {
127         course.setCourseName("");
128         course.setDescription("");
129         return;
130     }
131     Map<String, CourseTranslation> byLoc = new HashMap<>();
132     for (CourseTranslation r : rows) {
133         byLoc.put(r.getLocale(), r);
134     }
135     CourseTranslation chosen = byLoc.get(requestedLocale);
136     if (chosen == null) {
137         chosen = byLoc.get(course.getDefaultLocale());
138     }
139     if (chosen == null) {
140         chosen = rows.get(0);
141     }
142     course.setCourseName(chosen.getCourseName());
143     course.setDescription(chosen.getDescription() != null ? chosen.getDescription() : "");
144 }
145 olga.s... }
146

```

After resolving identified security issues and adding missing tests for services we got acceptable SonarQube analysis state.

Overview

[Set as homepage](#)

1.6k Lines of Code • Version not provided • Last analysis 48 seconds ago

New Code 1 failed

Overall Code

Security 0 Open issues A	Reliability 0 Open issues A	Maintainability 29 Open issues A
Accepted issues 0 Valid issues that were not fixed	Coverage 82.7% On 635 lines to cover.	Duplications 1.6% On 2.3k lines.
Security Hotspots 0 A		